NON-MF GROUPS

SAUERS

Some groups are not MF. We construct groups for which every homomorphism to an MF group is trivial; the obstruction is a one-sided compression of a property-(T) subgroup. If a countable unital ring R contains s, t with $ts = 1$ and $R(1 - st)R = R$, then every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial for all $n \geq 4$. In particular, $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ is a finitely generated simple property-(T) group that is not MF, and $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF. We also construct a sofic non-MF group $W = K \rtimes \mathbb{Z}$ with K locally residually finite, whose canonical trace on $C^*_{\max}(W)$ is amenable but not quasidiagonal, and a two-generated, finitely presented, torsion-free, acylindrically hyperbolic property-(T) group with no nontrivial homomorphism to an MF group.

Draft. This manuscript is under revision.

1. Introduction

In late July 2026 the author began using AI models on the question of whether every group is hyperlinear, inspired by the nonsofic group of [OAI]. In one ChatGPT conversation, GPT-5.6 Sol, on "high" thinking mode, worked on weak MF approximations of a Clifford-twisted wreath product as a step toward hyperlinearity. Earlier in that conversation the author had asked whether to pivot to the MF question, and Sol had advised against it. The author gave Sol the existing progress on the hyperlinear problem and sent this prompt:

> "Achieve f{MF of the Clifford Kun–Thom extension }$\widetilde{W}$ in such a way that it archives our goals and ends everything, or, possibly, a different breakthrough, leading to achieving the final result. You got this!"

Sol answered with an MF counterexample: every norm-matrix model of the Clifford-twisted wreath product kills the Clifford sign. That was on August 9. Also on August 9, the models proved that the maximal group C^*-algebra of a strictly compressed Kazhdan group is infinite. Parallel chats, including one in "Pro" mode, and the agents working on the hyperlinear question, did not find the counterexample. By August 12 the models had reduced the obstruction to one commutator in a finitely presented group and had proved it in Lean. They wrote the first draft of this paper that evening.

The author prompted the models, set the direction, retrieved the literature, suggested the MF question, and checked and revised the arguments and the citations. The proofs came from the models, over thousands of prompts. GPT-5.6 Sol in Codex, and Claude Fable 5 and Claude Opus 5 in Claude Code, wrote the code and the Lean proofs. The

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Lean proofs check the theorems, the assumptions, and the uses of the literature. A custom tool provided scaffolding for the informal mathematics, and a supercomputer ran computational experiments that ruled out preliminary approaches. The whole exchange is in the repository's public commit history.

The author consulted experts from August 13, and their comments shaped the revisions. Many models also reviewed drafts; Astra's reviews shortened the paper and simplified the proof of Theorem 2.7. Eckhardt also used ChatGPT to construct non-MF groups and wrote the expository note [Ec], a good first read on the compression mechanism. For the shortest route to a non-MF group, the reader may start there and return here for the compression criterion, the simple and torsion-free examples, and the amenable trace. The August construction is the group W of Theorem D, and Theorem A generalizes the argument for the central sign.

The counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)),$$

the elementary group of 12×12 matrices over the binary Leavitt algebra. Khanh–Thanh show that H is isomorphic to the unit group of the binary Leavitt algebra [K–T, Proposition 4.2 and Corollary 4.4].[1] OpenAI used this group to construct the first nonsofic group [OAI, Chapter 3].

We work with countable groups throughout. A countable group G is *MF* [CDE] if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

In other words, G is MF if there are finite unitary matrices, one for each element of G, that multiply correctly up to an error tending to zero in operator norm while no nonidentity element converges to the identity matrix. For countable G, the separation may equivalently be required along the full sequence with a constant independent of g [Kor, Propositions 2 and 7]. The strong convergence convention of [GKM, Sch] also requires the models to reproduce the operator norms of the left regular representation, so a group that is not MF as defined here is not MF in that convention either. Equivalently, G embeds in the unitary group of the norm matrix corona

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) \Big/ \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here and below, products of C^*-algebras mean bounded products, and $\bigoplus$ denotes the ideal of norm-null sequences. The quotient $\mathcal{Q}_{\mathbf{d}}$ is the corona algebra of $\bigoplus_n M_{d_n}(\mathbb{C})$, since $\prod_n M_{d_n}(\mathbb{C})$ is its multiplier algebra. We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. A separable C^*-algebra is MF if it embeds in a norm matrix corona [BK].

We use the group commutator convention $[g,h] = ghg^{-1}h^{-1}$. The groups here fail to be MF because they contain a property-(T) subgroup that is conjugated into a

[1] By the same results $\mathrm{GL}_n(R) = \mathrm{EL}_n(R) \cong R^\times$ for every $n \geq 2$, so every $\mathrm{EL}_n(R)$ is isomorphic to H; the rank 12 is the one checked in the Lean development.

proper subgroup of itself. For $L \le G$ put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \le L\}.$$

An element c of the centralizer $C_G(L)$ commutes with L, so ucu^{-1} commutes with uLu^{-1} but need not commute with the rest of L. The normal subgroup of G generated by the commutators of all such elements ucu^{-1} with elements of L is

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \le G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \le \mathfrak{D}_G(L)$, then every homomorphism from G to an MF group is trivial on K. In particular, a nontrivial such K makes G non-MF, and if G has property* (T) *and $\mathfrak{D}_G(L) = G$, then every homomorphism from G to an MF group is trivial.*

Asymptotic multiplicativity in operator norm makes the conjugation maps $\mathrm{Ad}(V_n(g))\colon x \mapsto V_n(g)xV_n(g)^*$ an asymptotically multiplicative family of unitaries on $M_{d_n}(\mathbb{C})$ with its normalized Hilbert–Schmidt inner product. Here $\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\|A - B\|$, with the norm on the left taken in the operator algebra of this Hilbert space. Passing to a Hilbert ultraproduct turns these maps into a genuine unitary representation of G, to which property (T) applies. Let U be the class of $(\mathrm{Ad}(V_n(u)))$ and P the Kazhdan projection of L for the conjugation maps, both in the norm matrix corona with coordinate sizes d_n^2. Then $U^*PU \le P$, because $uLu^{-1} \le L$ has fewer constraints and more fixed vectors, and stable finiteness of that corona forces $U^*PU = P$. So conjugation by u preserves the Hilbert–Schmidt asymptotic commutant of L, and every element of $\mathfrak{D}_G(L)$ tends to 1 in Hilbert–Schmidt norm in every operator norm asymptotic representation (Corollary 2.5). Property (T) of K turns this into triviality in operator norm. A corona homomorphism nontrivial on K compresses to a corner where the Kazhdan projection of K vanishes. On that corner, Hilbert–Schmidt triviality makes every ultralimit of normalized traces equal to the trivial character of K, which takes the value 1 at the Kazhdan projection (Theorem 2.7).[2]

The stable letter of an ascending HNN extension of a property-(T) group along a proper self-embedding is a compressor. For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \ne j$. A one-sided inverse $ts = 1$ in R is a compressor in the same way: an explicit element $u \in \mathrm{EL}_4(R)$ conjugates $\mathrm{EL}_3(R)$ into itself by $e_{ij}(a) \mapsto e_{ij}(sat)$, and if $1 - st$ generates R as a two-sided ideal, then one commutator $[ucu^{-1}, \ell]$ normally generates all of $\mathrm{EL}_n(R)$.

Theorem B (full complementary idempotents)**.** *Let R be a countable unital associative ring. Suppose that $s, t \in R$ satisfy*

$$ts = 1, \qquad R(1 - st)R = R.$$

For every $n \ge 4$, every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial.

[2] A Hilbert–Schmidt bound on the multiplicative defect does not give an operator norm bound on the conjugation maps, so this argument does not apply to sofic or hyperlinear approximations.

Here $R(1-st)R$ is the two-sided ideal generated by $1-st$, so the hypothesis says that $\sum_j a_j(1-st)b_j = 1$ for finitely many $a_j, b_j \in R$; we call such an idempotent $1-st$ *full*. A unital ring is *directly finite* if $ts = 1$ implies $st = 1$. The hypothesis holds in every countable simple unital ring that is not directly finite and in every Leavitt algebra $L_k(1,m)$ over a countable field k, $m \geq 2$ (Corollary 3.1), and R need not be finitely generated. Simplicity of $\mathrm{EL}_n(R)$ is a separate matter; it holds in the following example. Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra, the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

Theorem C (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is finitely generated, nontrivial, simple, has property* (T)*, and every homomorphism from H to an MF group is trivial. So H is not MF, and $C^*_{\mathrm{r}}(H)$ is separable and stably finite but not MF, while $C^*_{\max}(H)$ is not even finite: it contains a proper isometry.*

Theorem D (an amenable nonquasidiagonal trace)**.** *There is a sofic group $W = K \rtimes \mathbb{Z}$, with K locally residually finite, that is not MF. The canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal.*

Section 4 constructs W from the affine group $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ and Clifford lamps. The group W is sofic because locally residually finite groups are sofic and soficity passes to extensions with amenable quotient [ES, Theorem 1]. The trace answers the question of Brown [Br, discussion preceding Proposition 3.5.1] and of Schafhauser, Tikuisis and White [STW, Problem X(1)] whether every amenable trace is quasidiagonal. Since K is a direct limit of residually finite groups, it is MF [Kor, Corollary 10 and Proposition 13]. So MF groups are not closed under semidirect products with $\mathbb{Z}$, which answers a question of Korchagin [Kor, remark following Proposition 12].

Theorem E (a torsion-free finitely presented example)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group Q with property* (T) *such that every homomorphism from Q to an MF group is trivial. In particular, no nontrivial quotient of Q is MF.*

Section 5 builds Q from Fournier-Facio's torsion-free property-(T) group [FF, §2] and Hull's small cancellation theorem [Hu].

Related work. Blackadar and Kirchberg introduced MF algebras and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. Whether every separable stably finite C^*-algebra is MF, the MF problem [GH, §6], was answered negatively through the failure of the Connes embedding problem [JNVWY][GH, Proposition 6.1 and Remark 6.2]. Theorem C gives a counterexample among reduced group C^*-algebras. Whether every countable group is MF was a longstanding open problem [Kor, BDL].

Theorem A uses the configuration of the nonsofic construction [OAI, Proposition 2.3]: a property-(T) subgroup L conjugated into itself by an element u, and an element c commuting with L whose conjugate ucu^{-1} does not. There, soficity and the rigidity theorems of Kun [Ku16] and Kun–Thom [KT19] force the centralizing subgroup to be locally embeddable into finite groups, and a copy of Thompson's group V gives the contradiction. Here the rigidity is the Kazhdan projection of L in a norm matrix corona, which stable finiteness of the corona forces to commute with the compressor, and the contradiction is a commutator $[ucu^{-1}, \ell] \neq 1$ that every operator norm asymptotic representation sends to 1 in Hilbert–Schmidt norm. The group H is isomorphic to the unit group of the binary Leavitt algebra, the group of [OAI], and Theorem E reuses Fournier-Facio's configuration [FF]. Bachner–Dogon–Lubotzky showed that for groups of Deligne type, operator–Hilbert–Schmidt stability would imply that the group is not MF [BDL, Proposition 1.5]. Here the compression relation replaces the stability hypothesis.

Leavitt introduced the algebras $L_K(1,n)$ [Lev]. The binary Leavitt algebra is purely infinite simple [AA], is an exchange ring [Ara], and has center $\mathbb{F}_2$ [AC]. Preusser's sandwich theorem then gives simplicity of H [Pre], and property (T) is Ershov–Jaikin-Zapirain's theorem [EJZ, Theorem 1.1].

2. One-sided compression

2.1. Corona homomorphisms. If G is countable, the image of a corona homomorphism from G is a countable subgroup of the unitary group of a norm matrix corona, so it is MF. Every MF group embeds in the unitary group of a norm matrix corona.

Lemma 2.1. *Let G be countable and $K \leq G$. Every homomorphism from G to an MF group is trivial on K if and only if every corona homomorphism from G is trivial on K.*

Proof. The image of a corona homomorphism is a countable MF group. Conversely, compose a homomorphism to an MF group with a corona embedding of its image; the resulting corona homomorphism has the same kernel. □

2.2. Kazhdan transport in normalized Hilbert–Schmidt norm. For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a group G is a sequence of maps $V_n \colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

The elements g with $\|V_n(g) - 1\|_2 \to 0$ form a normal subgroup of G, because $\|a\|_2 \leq \|a\|$, the Hilbert–Schmidt norm is invariant under adjoints and unitary conjugation, and $\|V_n(g^{-1}) - V_n(g)^*\| \to 0$:

$$\|V_n(gh) - 1\|_2 \leq \|V_n(g) - 1\|_2 + \|V_n(h) - 1\|_2 + o(1),$$

$$\left\|V_n(g^{-1}) - 1\right\|_2 = \|V_n(g) - 1\|_2 + o(1), \qquad \left\|V_n(ghg^{-1}) - 1\right\|_2 = \|V_n(h) - 1\|_2 + o(1).$$

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|V_n(\ell)x_n - x_nV_n(\ell)\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.2. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C})/\bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Write A for the displayed corona. If $v^*v = 1$ in A and (x_n) is a bounded lift of v, then $\|x_n^*x_n - 1\| \to 0$, so for all large n the matrix $u_n = x_n(x_n^*x_n)^{-1/2}$ is unitary and $\|u_n - x_n\| \to 0$; after assigning arbitrary unitary values to the finitely many remaining coordinates, (u_n) is a unitary lift of v, and $vv^* = 1$. So A is finite, and so is $M_k(A)$, which is the corona with matrix sizes km_n. Finally, if $p \le q$ are equivalent projections in a finite algebra B and w is a partial isometry with $w^*w = q$ and $ww^* = p$, then $w = qwq$, so $w + (1-q)$ is an isometry of B and hence unitary; therefore $p + (1-q) = (w+(1-q))(w+(1-q))^* = 1$ and $p = q$. □

For a group L with property (T), the Kazhdan projection is the central projection $e_L \in C^*_{\max}(L)$ whose image in every unitary representation of L is the orthogonal projection onto the L-invariant vectors [AW].

Lemma 2.3 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Let $P \in B$ be the image of e_L under the $*$-homomorphism $C^*_{\max}(L) \to B$ induced by π. If a unitary $U \in B$ satisfies $U\pi(L)U^* \subseteq \pi(L)$, then $U^*PU \le P$.*

Proof. Fix a faithful nondegenerate representation of B on a Hilbert space $\mathcal{H}$, and let $\operatorname{Fix}\pi(L)$ be the space of vectors fixed by $\pi(L)$. Then P is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If $\xi \in \operatorname{Fix}\pi(L)$ and $\ell \in L$, then $\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi$, because $U\pi(\ell)U^*$ belongs to $\pi(L)$. So $U^*\operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$. Since U^*PU is the orthogonal projection onto $U^*\operatorname{Fix}\pi(L)$, it follows that $U^*PU \le P$. □

Theorem 2.4 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\mathrm{Ad}(V(u))(\mathcal{C}_2(V,L)) = \mathcal{C}_2(V,L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V,L)$ and a free ultrafilter ω, and let $\mathcal{K}_\omega$ be the ultraproduct of the Hilbert spaces $(M_{d_n}(\mathbb{C}), \|\cdot\|_2)$. For unitaries A, B,

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|\,,$$

where the norm on the left is the operator norm on this Hilbert space, so the maps $\mathrm{Ad}(V_n(g))$ are asymptotically multiplicative in operator norm, and

$$\sigma(g) = \lim_\omega \mathrm{Ad}(V_n(g))$$

is a unitary representation of G on $\mathcal{K}_\omega$ with $\xi = [x_n]_\omega \in \operatorname{Fix}\sigma(L)$. The same estimate makes

$$\widetilde{\sigma}(g) = [\operatorname{Ad}(V_n(g))]_n \in \mathcal{U}(\mathcal{B}), \qquad \mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) \Big/ \bigoplus_n B(M_{d_n}(\mathbb{C})),$$

a homomorphism, where $B(M_{d_n}(\mathbb{C}))$ denotes the operators on the Hilbert–Schmidt Hilbert space. Coordinatewise action induces a representation $\pi_\omega \colon \mathcal{B} \to B(\mathcal{K}_\omega)$, which we suppress from the notation. Moreover, $\mathcal{B}$ is a norm matrix corona with coordinate sizes d_n^2 after a choice of matrix units. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection of L [AW] under $C^*_{\max}(L) \to \mathcal{B}$. On $\mathcal{K}_\omega$ its range is $\operatorname{Fix}\sigma(L)$. Put $U = \widetilde{\sigma}(u)$. Since $uLu^{-1} \le L$,

$$U\widetilde{\sigma}(L)U^* \subseteq \widetilde{\sigma}(L), \qquad \text{so} \qquad U^*PU \le P$$

by Lemma 2.3, and

$$U^*PU = P$$

by Lemma 2.2, since the two projections are unitarily equivalent. So U commutes with P, and

$$U\xi,\ U^*\xi \in \operatorname{Fix}\sigma(L), \qquad \text{that is,}$$

$$\lim_\omega \left\| V_n(\ell)V_n(u)^{\pm 1}x_nV_n(u)^{\mp 1} - V_n(u)^{\pm 1}x_nV_n(u)^{\mp 1}V_n(\ell) \right\|_2 = 0 \qquad (\ell \in L).$$

Since ω was arbitrary, the limits hold along the full sequence: if one of these norms stayed above some $\varepsilon > 0$ on an infinite set of indices, a free ultrafilter containing that set would give a nonzero ultralimit. So $\operatorname{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$. ☐

Corollary 2.5. *Let $L \le G$ have property* (T) *and let (V_n) be an operator norm asymptotic representation of G. Then*

$$\|V_n(d) - 1\|_2 \longrightarrow 0 \qquad (d \in \mathfrak{D}_G(L)).$$

Proof. Let $u \in \operatorname{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, and write $\mathcal{C}_2 = \mathcal{C}_2(V, L)$. Since c commutes with L, $(V_n(c)) \in \mathcal{C}_2$, so $(V_n(u)V_n(c)V_n(u)^*) \in \mathcal{C}_2$ by Theorem 2.4, and then $(V_n(ucu^{-1})) \in \mathcal{C}_2$ because $\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \to 0$. So $\|V_n(\ell)V_n(ucu^{-1}) - V_n(ucu^{-1})V_n(\ell)\|_2 \to 0$, and by asymptotic multiplicativity $\|V_n([ucu^{-1}, \ell]) - 1\|_2 \to 0$. The elements with this property form a normal subgroup of G, so it contains $\mathfrak{D}_G(L)$.[3] ☐

[3]For a finite-dimensional unitary representation $\rho \colon G \to \mathcal{U}(d)$ the same conclusion holds without property (T). If $uLu^{-1} \le L$, then conjugation by $\rho(u)^*$ is an injective linear map of the commutant $\rho(L)'$ into itself, so it is onto. For $c \in C_G(L)$ the element $\rho(ucu^{-1}) = \rho(u)\rho(c)\rho(u)^*$ therefore lies in $\rho(L)'$, and ρ is trivial on $\mathfrak{D}_G(L)$. In a norm matrix corona this dimension count is not available, and Theorem 2.4 uses property (T) and stable finiteness in its place.

2.3. **From Hilbert–Schmidt to operator norm.** A corona homomorphism ρ cannot simply be restricted to a corner: a projection q commuting with $\rho(G)$ has lifts q_n that only asymptotically commute with chosen unitary lifts $U_n(g)$ of $\rho(g)$, so the compressions $q_nU_n(g)q_n$ are only approximately unitary in the matrix corners.

Lemma 2.6 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \operatorname{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$ such that, in the corona over the retained coordinates, the class of $(J_nW_n(g)J_n^*)$ is the coordinate restriction of $q\rho(g)$, a unitary of the corner $q\mathcal{Q}_{\mathbf{d}}q$ with unit q.

Proof. Lift q to projections q_n by spectral rounding, and lift each $\rho(g)$ to unitaries $U_n(g)$ by polar correction, with $U_n(1) = I_{d_n}$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain those coordinates and put $r_n = \operatorname{rank}(q_n)$. Since q commutes with $\rho(G)$, $\|q_nU_n(g) - U_n(g)q_n\| \to 0$ for each g, so by [BDL, Proposition 2.4] with $p = \infty$ there is an operator norm asymptotic representation $\widetilde{W}_n$ of G by unitaries of the corner $q_nM_{d_n}(\mathbb{C})q_n$ with $\left\|\widetilde{W}_n(g) - q_nU_n(g)q_n\right\| \to 0$. Replacing $\widetilde{W}_n(1)$ by q_n changes it by $o(1)$. Choose unitaries $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$ and put $W_n(g) = J_n^*\widetilde{W}_n(g)J_n$. Conjugation by J_n preserves operator norms and multiplicative defects, so (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$, and $\|J_nW_n(g)J_n^* - q_nU_n(g)q_n\| \to 0$ identifies the class of $(J_nW_n(g)J_n^*)$ with the coordinate restriction of $q\rho(g)$. ☐

Theorem 2.7 (normal Kazhdan subgroups)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T)*. If every operator norm asymptotic representation (V_n) of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for all $k \in K$, then every corona homomorphism from G is trivial on K.*

Proof. Suppose that a corona homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Let $e_K \in C^*_{\max}(K)$ be the Kazhdan projection of K, and let $p \in \mathcal{Q}_{\mathbf{d}}$ be its image under the homomorphism $C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{d}}$ induced by $\Theta|_K$. In any faithful representation of the corona, p is the projection onto the K-fixed vectors. Since $gKg^{-1} = K$, the range of $\Theta(g)p\Theta(g)^*$ is $\Theta(g)\operatorname{Fix}\Theta(K) = \operatorname{Fix}\Theta(gKg^{-1}) = \operatorname{Fix}\Theta(K)$, so

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G),$$

and $q = 1 - p$ is nonzero, because $p = 1$ would make every element of K act trivially. Lemma 2.6 gives an operator norm asymptotic representation $W_n\colon G \to \mathcal{U}(r_n)$ whose corona homomorphism $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$. Let $\pi\colon C^*_{\max}(K) \to \mathcal{Q}_{\mathbf{r}}$ be the homomorphism induced by $\widehat{\Theta}|_K$. Since q commutes with $\Theta(K)$, the map $a \mapsto q\Theta(a)$ is a homomorphism on $C^*_{\max}(K)$ agreeing with π on K, so $\pi(e_K)$ is the coordinate restriction of $qp = 0$.

Fix a free ultrafilter ω. Since $|\operatorname{tr}_{r_n}(a_n)| \le \|a_n\|$, the formula

$$T_\omega([a_n]) = \lim_\omega \operatorname{tr}_{r_n}(a_n)$$

defines a tracial state on $\mathcal{Q}_{\mathbf{r}}$. The hypothesis applies to (W_n) itself, so

$$|\operatorname{tr}_{r_n}(W_n(k)) - 1| \le \|W_n(k) - I_{r_n}\|_2 \longrightarrow 0 \qquad (k \in K),$$

with the normalized Hilbert–Schmidt norm of $M_{r_n}(\mathbb{C})$, and therefore $T_\omega(\pi(k)) = 1$ for every $k \in K$. Then the states $T_\omega \circ \pi$ and the trivial character of K agree on the canonical unitaries, whose span is dense in $C^*_{\max}(K)$, and therefore on all of $C^*_{\max}(K)$. The trivial character takes the value 1 at e_K, while $\pi(e_K) = 0$. So $0 = T_\omega(\pi(e_K)) = 1$, a contradiction. □

Proof of Theorem A. By Corollary 2.5, every operator norm asymptotic representation of G satisfies $\|V_n(k) - 1\|_2 \to 0$ for $k \in K \le \mathfrak{D}_G(L)$. By Theorem 2.7, every corona homomorphism is then trivial on K, and by Lemma 2.1 so is every homomorphism to an MF group. If $K \ne 1$, then G does not embed in an MF group, so G is not MF. The last assertion is the case $K = G = \mathfrak{D}_G(L)$. □

2.4. **The maximal group C^*-algebra.** Recall that a unital C^*-algebra is finite if each of its isometries is unitary, and stably finite if all matrix algebras over it are finite.

Lemma 2.8 (strictly dominated projections)**.** *Let A be a unital C^*-algebra, let $p \in A$ be a projection, and let $u \in A$ be a unitary with $p < upu^*$. Then $s = pu^* + (1 - upu^*)$ is an isometry with $ss^* \ne 1$. In particular A is not stably finite, and no tracial state on A is faithful.*

Proof. Put $q = upu^*$, so that $pq = qp = p$, $up(1-q) = 0$, and $pu^*(1-q) = pu^* - pu^*upu^* = 0$. Then

$$\begin{aligned} s^*s &= upu^* + up(1-q) + (1-q)pu^* + (1-q) = q + (1-q) = 1, \\ ss^* &= pu^*up + pu^*(1-q) + (1-q)up + (1-q) \\ &= p + (1-q) = 1 - (q-p) \ne 1. \end{aligned}$$

A unital C^*-algebra containing a nonunitary isometry is not finite, and neither is any matrix algebra over it, since $\operatorname{diag}(s, 1, \ldots, 1)$ is a nonunitary isometry in $M_k(A)$. A tracial state τ has $\tau(q-p) = \tau(s^*s) - \tau(ss^*) = 0$ with $q - p$ a nonzero positive element, so no tracial state on A is faithful. □

Proposition 2.9 (proper one-sided compression)**.** *Let G be a countable group containing a property-(T) subgroup Γ and an element t with $t\Gamma t^{-1} \subsetneq \Gamma$. Then $C^*_{\max}(G)$ contains a proper isometry. In particular, $C^*_{\max}(G)$ is not stably finite, admits no faithful tracial state, and is neither residually finite-dimensional nor MF.*

Proof. Let $P \in C^*_{\max}(G)$ be the image of the Kazhdan projection of Γ [AW] under the canonical map $C^*_{\max}(\Gamma) \to C^*_{\max}(G)$, and let u be the canonical unitary of t. Conjugation by u implements the isomorphism $\Gamma \to t\Gamma t^{-1}$ on canonical unitaries, so by Lemma 2.3 applied with $U = u$, $P \le uPu^*$, where uPu^* is the image of the Kazhdan projection of $t\Gamma t^{-1}$. The inequality is strict: in the quasi-regular representation of G

on $\ell^2(G/t\Gamma t^{-1})$, the point mass at the base coset is fixed by $t\Gamma t^{-1}$ and moved by every element of $\Gamma \setminus t\Gamma t^{-1}$, so the two fixed-space projections differ. By Lemma 2.8, $C^*_{\max}(G)$ contains a proper isometry, is not stably finite, and has no faithful tracial state. MF algebras are stably finite, and a separable residually finite-dimensional C^*-algebra is MF [BK]. □

3. One-sided inverses and elementary groups

The proof of Theorem B uses four coordinates: three for the property-(T) subgroup $\mathrm{EL}_3(R)$ and one for its centralizer. The compressor needs only $ts = 1$; the ideal condition on $1 - st$ enters in the final normal-generation step. Indices run over $1, \dots, n$ throughout this section.

Proof of Theorem B. Write $e = 1 - st$. Then $e^2 = e$, $es = te = 0$, and there are finitely many $a_j, b_j \in R$ with $\sum_j a_j e b_j = 1$. First suppose that R is finitely generated as a unital ring. Put $G = \mathrm{EL}_n(R)$ and let $L = \mathrm{EL}_3(R)$ occupy coordinates $1, 2, 3$. Both groups have property (T) by [EJZ, Theorem 1.1]. All displayed 4×4 matrices below are extended by I_{n-4}.

For $i = 1, 2, 3$, set

$$u_i = e_{4i}(t-1)e_{i4}(1)e_{4i}(s-1)e_{i4}(-t).$$

Its block on coordinates $(i, 4)$ is $\left(\begin{smallmatrix} s & e \\ 0 & t \end{smallmatrix}\right)$, so $u = u_3u_2u_1 \in \mathrm{EL}_4(R)$ is the first of the two matrices

$$u = \begin{pmatrix} s & 0 & 0 & e \\ 0 & s & 0 & et \\ 0 & 0 & s & et^2 \\ 0 & 0 & 0 & t^3 \end{pmatrix}, \qquad u^{-1} = \begin{pmatrix} t & 0 & 0 & 0 \\ 0 & t & 0 & 0 \\ 0 & 0 & t & 0 \\ e & se & s^2e & s^3 \end{pmatrix}.$$

In block form $u = \left(\begin{smallmatrix} sI_3 & v \\ 0 & t^3 \end{smallmatrix}\right)$ with $v = (e, et, et^2)^{\mathsf{T}}$. Put $w = (e, se, s^2e)$. Since $es = te = 0$, one has $vw = eI_3$, $vs^3 = 0$, and $t^3w = 0$, so the second matrix is the inverse of u, and for $A \in M_3(R)$

$$u\,\mathrm{diag}(A, 1)u^{-1} = \begin{pmatrix} sAt + vw & vs^3 \\ t^3w & t^3s^3 \end{pmatrix} = \mathrm{diag}(eI_3 + sAt, 1).$$

In particular $ue_{ij}(a)u^{-1} = e_{ij}(sat)$ for $1 \le i \ne j \le 3$, because $e + st = 1$, so $uLu^{-1} \le L$.

The element

$$c = [e_{41}(e), e_{14}(t)] = \mathrm{diag}(1, 1, 1, 1 + et)$$

is computed in the block on coordinates $(1, 4)$ using $te = 0$, and a diagonal matrix of this shape commutes with every $\mathrm{diag}(A, 1)$, so $c \in C_G(L)$. Since $t^3e = 0$, the matrix uc differs from u only in its last column, which is $(e + et, et, et^2, t^3)^{\mathsf{T}}$, and multiplying by u^{-1} gives

$$ucu^{-1} = e_{12}(e).$$

For $\ell = e_{23}(1)$ the Steinberg relation $[e_{12}(e), e_{23}(1)] = e_{13}(e)$ gives

$$d = [ucu^{-1}, \ell] = e_{13}(e) \in \mathfrak{D}_G(L).$$

Let N be the normal closure of d in G. For arbitrary $a, b \in R$, the Steinberg relations give

$$[e_{41}(a), e_{13}(e)] = e_{43}(ae), \qquad [e_{43}(ae), e_{32}(b)] = e_{42}(aeb).$$

It follows that $e_{42}(1) = \prod_j e_{42}(a_j e b_j) \in N$. Elementary signed permutation matrices conjugate this root to every root position, up to a sign, so $e_{ij}(1) \in N$ whenever $i \neq j$. For distinct i, j, k and every $r \in R$,

$$[e_{ij}(1), e_{jk}(r)] = e_{ik}(r) \in N.$$

Hence $N = G$, and therefore $\mathfrak{D}_G(L) = G$. By Theorem A with $K = G$, every homomorphism from G to an MF group is trivial.

For general countable R, every element of $\mathrm{EL}_n(R)$ is a product of finitely many elementary matrices, so it lies in $\mathrm{EL}_n(S)$ for some finitely generated unital subring S of R containing s, t, and the a_j, b_j. The hypotheses hold in S, so a homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial on each such $\mathrm{EL}_n(S)$, and these subgroups cover $\mathrm{EL}_n(R)$.[4] □

Corollary 3.1. *If R is a countable simple unital ring that is not directly finite, then every homomorphism from $\mathrm{EL}_n(R)$ to an MF group is trivial for every $n \geq 4$. The same conclusion holds for $R = L_k(1, m)$, for every countable field k and every $m \geq 2$.*

Proof. In the first case choose $ts = 1 \neq st$; then $1 - st$ is a nonzero idempotent, and since R is simple it generates R as a two-sided ideal. In the second case, with generators $s_1, \dots, s_m, t_1, \dots, t_m$ subject to $t_i s_j = \delta_{ij}$ and $\sum_i s_i t_i = 1$, take $s = s_1$ and $t = t_1$; then $1 - s_1 t_1 = \sum_{i \geq 2} s_i t_i$ and

$$t_2(1 - s_1 t_1) s_2 = 1.$$

Theorem B applies in both cases. □

Corollary 3.2. *If a countable unital ring R is not directly finite, then $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry for every $n \geq 4$. If $R \neq 0$ also satisfies the hypothesis of Theorem B, then $C^*_{\mathrm{r}}(\mathrm{EL}_n(R))$ is separable, stably finite, and not MF.*

Proof. Choose $s, t \in R$ with $ts = 1 \neq st$, and let S be the unital subring they generate. The compressor u from the proof of Theorem B lies in $\mathrm{EL}_4(S)$ and conjugates $L = \mathrm{EL}_3(S)$ into itself, and L has property (T) [EJZ, Theorem 1.1]. The compression is strict: every off-diagonal entry of a matrix in uLu^{-1} has the form sat, and $e(sat) = 0$ while $e \cdot 1 = e \neq 0$, so $e_{12}(1) \in L \setminus uLu^{-1}$. By Proposition 2.9 applied to $L \leq \mathrm{EL}_n(R)$ and u, $C^*_{\max}(\mathrm{EL}_n(R))$ contains a proper isometry.

For the second assertion, $G = \mathrm{EL}_n(R)$ is countable and nontrivial, and by Theorem B it is not MF. The algebra $C^*_{\mathrm{r}}(G)$ is separable, and stably finite because its canonical trace is faithful. If an embedding $\iota\colon C^*_{\mathrm{r}}(G) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = \iota(1)$, then $g \mapsto \iota(\lambda_g) + (1 - p)$ would embed G in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, contradicting that G is not MF. □

[4] The same argument applies when finitely many pairs $t_\nu s_\nu = 1$ have complementary idempotents $1 - s_\nu t_\nu$ that together generate the unit ideal: each pair gives a compressor and a centralizer for the same L, so each $e_{13}(1 - s_\nu t_\nu)$ lies in $\mathfrak{D}_G(L)$, and the two Steinberg relations above put $e_{42}(1)$ in their normal closure.

3.1. **The binary example.** Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. By (2), the maps $x \mapsto (t_0x, t_1x)$ and $(y,z) \mapsto s_0y + s_1z$ are mutually inverse isomorphisms of right R-modules between R and $R \oplus R$. With $s = s_0$ and $t = t_0$, the relations give $t_0s_0 = 1$ and

$$t_1(1 - s_0t_0)s_1 = 1,$$

so R satisfies the hypothesis of Theorem B, and every homomorphism from H to an MF group is trivial. It remains to prove that H is simple.

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. Since $e_{12}(1) \neq 1$, the group H is nontrivial. Let N be a normal subgroup of H. The ring R is purely infinite simple [AA] and therefore an exchange ring [Ara], so by Preusser's sandwich theorem [Pre, Theorem 3] there is an ideal I of R with

$$\mathrm{EL}_{12}(R,I) \leq N \leq C_{12}(R,I),$$

where $\mathrm{EL}_{12}(R,I)$ is the normal closure in $\mathrm{EL}_{12}(R)$ of the elementary matrices $e_{ij}(a)$ with $a \in I$, and $C_{12}(R,I)$ is the preimage of the center of $\mathrm{GL}_{12}(R/I)$. Since R is simple, either $I = R$, and then $N \geq \mathrm{EL}_{12}(R) = H$, or $I = 0$, and then every $g \in N$ is central in $\mathrm{GL}_{12}(R)$. In the second case, commuting with each $e_{ij}(1)$ forces $g = \lambda I_{12}$ with $\lambda \in R^\times$, and commuting with each $e_{ij}(a)$ then gives $\lambda a = a\lambda$ for all $a \in R$. So λ lies in $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], and $N = 1$. □

Proof of Theorem C. The ring R is finitely generated, so H has property (T) by [EJZ, Theorem 1.1], and hence is finitely generated [BHV, Theorem 1.3.1]. By Theorem B, every homomorphism from H to an MF group is trivial, and by Proposition 3.3, H is nontrivial and simple. In particular H is not MF. Since $t_0s_0 = 1$ and $s_0t_0 \neq 1$, the ring R is not directly finite, so by Corollary 3.2 the algebra $C^*_{\mathrm{r}}(H)$ is separable, stably finite, and not MF, and $C^*_{\max}(H)$ contains a proper isometry. □

4. An amenable nonquasidiagonal trace

Brown observed that it was unknown whether every amenable trace on a C^*-algebra is quasidiagonal [Br, discussion preceding Proposition 3.5.1]. For a separable unital C^*-algebra A we use the sequential form of his definitions. A tracial state τ on A is *amenable* if there are u.c.p. maps $\phi_n \colon A \to M_{d_n}(\mathbb{C})$ such that

$$\|\phi_n(ab) - \phi_n(a)\phi_n(b)\|_2 \longrightarrow 0, \qquad \mathrm{tr}_{d_n}(\phi_n(a)) \longrightarrow \tau(a) \quad (a,b \in A).$$

It is *quasidiagonal* if the first limit holds in operator norm instead of normalized Hilbert–Schmidt norm, with the same trace convergence.

For a countable group G, let u_g denote the canonical unitary in $C^*_{\max}(G)$. The canonical trace τ_G is determined by

$$\tau_G(u_g) = \begin{cases} 1, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Theorem 4.1 (canonical-trace obstruction)**.** *Let G be a countable group. If G is not MF, then its canonical trace τ_G on $C^*_{\max}(G)$ is not quasidiagonal. In particular, if τ_G is amenable, then τ_G is an amenable trace that is not quasidiagonal.*

Proof. Suppose that τ_G is quasidiagonal, and let ϕ_n be u.c.p. maps as in the definition, with the first limit in operator norm. Then $\phi_n(u_g)^*\phi_n(u_g)$ and $\phi_n(u_g)\phi_n(u_g)^*$ converge to 1 in operator norm, so for large n the unitary part $V_n(g)$ of the polar decomposition of $\phi_n(u_g)$ satisfies

$$\|V_n(g) - \phi_n(u_g)\| \longrightarrow 0,$$

and $V_n(1) = 1$ because ϕ_n is unital. The maps $V_n \colon G \to \mathcal{U}(d_n)$ are asymptotically multiplicative in operator norm. If $g \neq 1$, then $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to \tau_G(u_g) = 0$, so $V_n(g)$ does not converge to 1 in operator norm, which would force $\mathrm{tr}_{d_n}(\phi_n(u_g)) \to 1$. So G is MF, contrary to the hypothesis. The last assertion follows. $\square$

A group N is *locally residually finite* if each of its finitely generated subgroups is residually finite.

Proposition 4.2 (extensions of $\mathbb{Z}$ by locally residually finite groups)**.** *Suppose that G is countable and*

$$G \cong N \rtimes_\beta \mathbb{Z},$$

where N is locally residually finite. Then τ_G is amenable.

Proof. Write $g = x_g t^{k_g}$, where $x_g \in N$, $k_g \in \mathbb{Z}$, and $t = (1, 1)$, so $txt^{-1} = \beta(x)$ for $x \in N$. Call k_g the height of g. Fix a finite set $F \subseteq G$ and a positive integer L, and put $I_L = \{0, \dots, L-1\}$. Define

$$a_{g,j} = t^{-(j+k_g)} g t^j = \beta^{-(j+k_g)}(x_g) \in N, \qquad g \in F, \quad j \in I_L,$$

and put $N_{F,L} = \langle a_{g,j} : g \in F,\ j \in I_L \rangle$. This finitely generated subgroup of N is residually finite, so there is a finite quotient

$$\theta_{F,L} \colon N_{F,L} \longrightarrow Q_{F,L}$$

that separates every nonidentity $a_{g,j}$: take the product of finitely many finite quotients, one separating each $a_{g,j}$, and restrict to the image. Let $H_{F,L} \leq G$ be the image of $\ker \theta_{F,L} \leq N_{F,L}$ under the inclusion $N \hookrightarrow G$. Choose a representative $r_q \in N_{F,L}$ for each $q \in Q_{F,L}$ and set

$$S_{F,L} = \{t^j r_q H_{F,L} : j \in I_L,\ q \in Q_{F,L}\} \subseteq G/H_{F,L}.$$

These cosets are pairwise distinct. Let $\lambda_{F,L}$ be the quasi-regular representation of G on $\ell^2(G/H_{F,L})$ and let $P_{F,L}$ be the projection onto $\ell^2(S_{F,L})$. Put

$$\phi_{F,L}(b) = P_{F,L}\lambda_{F,L}(b)P_{F,L}\big|_{\ell^2(S_{F,L})} \qquad (b \in C^*_{\max}(G)).$$

This map is u.c.p., being the compression of a unitary representation. For $g \in F$ and $j + k_g \in I_L$, the action satisfies

$$g(t^j r_q H_{F,L}) = t^{j+k_g} a_{g,j} r_q H_{F,L} = t^{j+k_g} r_{\theta_{F,L}(a_{g,j})q} H_{F,L}.$$

So for $s = t^j r_q H_{F,L}$, the point $g \cdot s$ lies in $S_{F,L}$ if and only if $j + k_g \in I_L$.

Let $g, h \in F$, and let δ_s be the basis vector of $\ell^2(S_{F,L})$ at $s = t^j r_q H_{F,L}$. If $j + k_h \in I_L$, then $h \cdot s \in S_{F,L}$, so $\phi_{F,L}(u_g)\phi_{F,L}(u_h)\delta_s = P_{F,L}\delta_{gh\cdot s} = \phi_{F,L}(u_{gh})\delta_s$. If $j + k_h \notin I_L$, then $h \cdot s \notin S_{F,L}$, so $\phi_{F,L}(u_h)\delta_s = 0$, while $\phi_{F,L}(u_{gh})\delta_s$ has norm at most one. There are at most $|k_h|$ heights $j \in I_L$ with $j + k_h \notin I_L$, and there are $|Q_{F,L}|$ basis vectors of each height, out of $L|Q_{F,L}|$ in all. So the normalized Hilbert–Schmidt norm satisfies

$$\|\phi_{F,L}(u_{gh}) - \phi_{F,L}(u_g)\phi_{F,L}(u_h)\|_2^2 \leq \frac{|k_h|}{L},$$

which tends to zero as $L \to \infty$ for fixed h.

The normalized trace of $\phi_{F,L}(u_g)$ is the fraction of points of $S_{F,L}$ fixed by g. If g fixed $wH_{F,L}$, then $w^{-1}gw \in H_{F,L} \leq N$, and hence $k_g = 0$. Thus an element of nonzero height fixes no point of $S_{F,L}$. If $g \in F \setminus \{1\}$ and $k_g = 0$, fixing $t^j r_q H_{F,L}$ would give

$$r_q^{-1} a_{g,j} r_q \in H_{F,L}.$$

Since $H_{F,L} = \ker \theta_{F,L}$ is normal in $N_{F,L}$ and $r_q \in N_{F,L}$, this implies $a_{g,j} \in \ker \theta_{F,L}$. But $a_{g,j} = t^{-j} g t^j \neq 1$, contrary to separation. Thus

$$\mathrm{tr}_{L|Q_{F,L}|}(\phi_{F,L}(u_g)) = 0 \quad (g \in F \setminus \{1\}), \qquad \mathrm{tr}_{L|Q_{F,L}|}(\phi_{F,L}(1)) = 1.$$

Take a finite exhaustion $F_1 \subseteq F_2 \subseteq \cdots$ of G with $1 \in F_1$, and choose $L_n \geq n(1 + \max_{g \in F_n} |k_g|)$. The maps ϕ_{F_n,L_n} are asymptotically multiplicative in normalized Hilbert–Schmidt norm and their traces converge to τ_G on every canonical unitary. Since the canonical unitaries span a dense subspace of $C^*_{\max}(G)$ and all the maps involved are contractive, both limits hold for all $a, b \in C^*_{\max}(G)$. So τ_G is amenable. $\square$

We now construct W. For W to be non-MF, Γ may be any countable property-(T) group with a proper self-embedding; residual finiteness of Γ and finiteness of the index of the self-embedding are needed only for the trace (Proposition 4.4). Let Γ be a countable group with property (T), let $\alpha\colon \Gamma \to \Gamma$ be injective but not surjective, and choose $a \in \Gamma \setminus \alpha(\Gamma)$. Put

$$T_\alpha = \varinjlim(\Gamma \xrightarrow{\alpha} \Gamma \xrightarrow{\alpha} \cdots), \qquad V = T_\alpha \rtimes \langle t \rangle,$$

where $tgt^{-1} = \alpha(g)$ on the level-zero copy of Γ, so that V is the ascending HNN extension of Γ along α and $T_\alpha = \bigcup_{n \geq 0} t^{-n}\Gamma t^n$. Put $X = V/\Gamma$, the left-coset space. The Clifford lamp group $\mathrm{Cl}(X)$ is the group with generators ε and c_x, $x \in X$, and relations

$$\varepsilon^2 = c_x^2 = 1, \qquad [\varepsilon, c_x] = 1, \qquad [c_x, c_y] = \varepsilon \quad (x \neq y).$$

We check that $\varepsilon \neq 1$ by building a model. Fix a total order on X, let E be the $\mathbb{F}_2$-vector space of finitely supported functions $X \to \mathbb{F}_2$, and for $f, g \in E$ put $B(f,g) = \sum_{x>y} f(x)g(y)$, a finite sum. Then

$$(a,f)(b,g) = \big(a + b + B(f,g),\ f + g\big)$$

defines a group structure on $\mathbb{F}_2 \times E$: the product is associative because B is bilinear, so that either way of multiplying (a,f), (b,g), (c,h) has first coordinate $a + b + c + B(f,g) + B(f,h) + B(g,h)$; the identity is $(0,0)$; and $(a + B(f,f), f)$ is inverse to (a,f). In this group $(1,0)$ is a central involution, and each $(0,\delta_x)$ is an involution because $B(\delta_x,\delta_x) = 0$. For $x \neq y$ exactly one of $B(\delta_x,\delta_y)$ and $B(\delta_y,\delta_x)$ equals 1, so the commutator of $(0,\delta_x)$ and $(0,\delta_y)$ is $(1,0)$. So $\varepsilon \mapsto (1,0)$, $c_x \mapsto (0,\delta_x)$ extends to a homomorphism $\mathrm{Cl}(X) \to \mathbb{F}_2 \times E$, and $\varepsilon \neq 1$. The relations let us write every element of $\mathrm{Cl}(X)$ as

$$\varepsilon^a c_{x_1} \cdots c_{x_r}, \qquad a \in \mathbb{F}_2, \quad x_1 < \cdots < x_r,$$

and this word maps to $(a, \delta_{x_1} + \cdots + \delta_{x_r})$, so the expression is unique and the homomorphism is an isomorphism. In particular, for a finite subset $Y \subseteq X$, the subgroup generated by ε and the c_y with $y \in Y$ has order $2^{|Y|+1}$; so $\mathrm{Cl}(X)$ is countable and locally finite. The relations are invariant under permutations of X, so every permutation of X induces an automorphism of $\mathrm{Cl}(X)$ that permutes the c_x accordingly and fixes ε. In this way V acts on $\mathrm{Cl}(X)$ through its action on X, and we put

$$W = \mathrm{Cl}(X) \rtimes V. \tag{3}$$

Proposition 4.3 (Clifford obstruction from a proper self-embedding)**.** *The group W is not MF. More precisely, every homomorphism from W to an MF group kills ε.*

Proof. Identify Γ with its level-zero copy in $V \leq W$, and let c be the lamp at the root coset $\Gamma \in X$; it centralizes Γ, which fixes that coset. Then tct^{-1} is the lamp at $t\Gamma$ and $a(tct^{-1})a^{-1}$ is the lamp at $at\Gamma$. These cosets are distinct, because $t\Gamma = at\Gamma$ would mean $a \in t\Gamma t^{-1} = \alpha(\Gamma)$. So

$$[tct^{-1}, a(tct^{-1})a^{-1}] = \varepsilon \neq 1. \tag{4}$$

Set

$$x = tct^{-1}, \qquad y = axa^{-1}, \qquad d = [x, a].$$

Both x and y are involutions, and hence

$$d = xaxa^{-1} = xy, \qquad d^2 = (xy)^2 = [x, y] = \varepsilon.$$

Since $t \in \mathrm{Comp}_W(\Gamma)$, $c \in C_W(\Gamma)$ and $a \in \Gamma$, the element $d = [tct^{-1}, a]$ is one of the generators of $\mathfrak{D}_W(\Gamma)$. Thus

$$\langle \varepsilon \rangle \leq \mathfrak{D}_W(\Gamma).$$

The subgroup $\langle \varepsilon \rangle$ is central, so normal, and it is finite, so it has property (T). The group W is countable, since T_α, V, X, and $\mathrm{Cl}(X)$ are countable. So Theorem A applies with $L = \Gamma$ and $K = \langle \varepsilon \rangle$: every homomorphism from W to an MF group is trivial on ε. In particular, W is not MF. □

Proposition 4.4 (the locally residually finite case)**.** *Suppose in addition that Γ is residually finite and that $[\Gamma : \alpha(\Gamma)] < \infty$. Then*

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha,$$

where K is locally residually finite. Consequently W is sofic but not MF, and the canonical trace on $C^*_{\max}(W)$ is amenable and not quasidiagonal.

Proof. Reassociating the semidirect products yields

$$W \cong K \rtimes \mathbb{Z}, \qquad K = \mathrm{Cl}(X) \rtimes T_\alpha.$$

For $n \geq 0$, write $\Gamma_n = t^{-n}\Gamma t^n$ for the image of the nth copy of Γ in T_α. Successive telescope copies have finite relative index because $[\Gamma : \alpha(\Gamma)] < \infty$, so all the Γ_n are commensurable. In fact Γ is commensurated by V: every element of T_α lies in some telescope copy, and conjugation by t shifts the copies. Hence the stabilizer in Γ_n of every point $v\Gamma \in V/\Gamma$ has finite index, and every Γ_n-orbit in X is finite.

A finite subset of K involves only finitely many lamps c_x, and its T_α-coordinates lie in one Γ_n. Let Y be the union of the Γ_n-orbits of these finitely many x. Then Y is finite and Γ_n-invariant, and the finite subset is contained in

$$C_Y \rtimes \Gamma_n,$$

where $C_Y = \langle \varepsilon, c_y : y \in Y\rangle$ is finite by the normal form above. This semidirect product is residually finite. Indeed, an element with nontrivial Γ_n-component survives in a finite quotient of Γ_n. For a nontrivial element of C_Y, let J be the kernel of the action $\Gamma_n \to \operatorname{Aut}(C_Y)$. The quotient Γ_n/J is finite, and the finite semidirect product $C_Y \rtimes (\Gamma_n/J)$ retains that element. Therefore every finitely generated subgroup of K is residually finite.

Each finitely generated subgroup of K is residually finite and hence sofic. The group K is their directed union, and $W/K \cong \mathbb{Z}$ is amenable. Soficity passes to directed unions and to extensions with amenable quotient [ES, Theorem 1], so W is sofic. Proposition 4.3 shows that it is not MF. By Proposition 4.2 the canonical trace is amenable, and by Theorem 4.1 it is not quasidiagonal. $\square$

For a concrete instance, take

$$\bar\Gamma = \mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z}) = \left\{ \begin{pmatrix} A & v \\ 0 & 1 \end{pmatrix} : A \in \mathrm{SL}_3(\mathbb{Z}),\ v \in \mathbb{Z}^3 \right\} \le \mathrm{GL}_4(\mathbb{Z}).$$

It is residually finite, since reduction modulo a suitable integer separates any two distinct integral matrices, and it has property (T) [BHV, Example 1.7.4(i)]. Let

$$D = \operatorname{diag}(2,2,2,1), \qquad \alpha(g) = DgD^{-1},$$

so that $\alpha(v, A) = (2v, A)$. This is injective, and its image consists of the affine matrices whose translation coordinates are all even, so $[\bar\Gamma : \alpha(\bar\Gamma)] = 8$. Translation a by the first standard basis vector lies outside $\alpha(\bar\Gamma)$. So the preceding construction applies to $\bar\Gamma$, α and a.

Corollary 4.5 (the affine–Clifford trace)**.** *The affine–Clifford group W is sofic but not MF. Its canonical trace τ_W on $C^*_{\max}(W)$ is amenable but not quasidiagonal.*

Proof. By the properties of $\bar\Gamma$ and α just established, Proposition 4.4 applies to $\bar\Gamma$, α, and a. $\square$

Proof of Theorem D. By Proposition 4.4, $W \cong K \rtimes \mathbb{Z}$ with K locally residually finite, and the remaining assertions are Corollary 4.5. $\square$

5. A torsion-free finitely presented example

For an acylindrically hyperbolic group G, choose the generating set A provided by Hull [Hu, Theorem 3.12]. A subgroup is *suitable* if it acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup [Hu, Definition 1.4]. Hull states his small cancellation theorem [Hu, Theorem 7.1] with injectivity on a ball of $\Gamma(G, A)$; since every finite subset of G lies in some ball, the theorem takes the following form.

Theorem 5.1 (Hull's small cancellation theorem)**.** *Let G be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to A, and let $g_1, \dots, g_m \in G$ and a finite subset $\Omega \subseteq G$ be given. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is acylindrically hyperbolic, $\varphi|_\Omega$ is injective, $\varphi(g_i) \in \varphi(N)$ for all i, and every element of finite order in Q is the image of an element of the same order in G.*

Hull's proof treats $m = 1$ by passing to $G/\langle\!\langle r \rangle\!\rangle_G$ for one element r and the general case by induction on m [Hu, proof of Theorem 7.1], so $\ker\varphi$ is the normal closure of m elements and Q is finitely presented when G is.

Lemma 5.2 (saturation)**.** *Let G be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $\Omega \subseteq G$ be finite. Then there is a surjective homomorphism $\varphi\colon G \to Q$ such that Q is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $\varphi|_\Omega$ is injective, and $\varphi(N) = Q$.*

Proof. The subgroup N is infinite and normal, so it acts non-elementarily on $\Gamma(G, A)$ by Osin [Os, Lemma 7.1]; since G is torsion-free, N normalizes no nontrivial finite subgroup, so N is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], N contains two elements h_1, h_2 such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable, possibly with respect to a larger generating set $A' \supseteq A$, and Theorem 5.1 holds for A' as well, since every finite subset of G lies in a ball of $\Gamma(G, A')$. Apply Theorem 5.1 to N_0 with $g_1, \dots, g_m$ a finite generating set of G and with the set Ω. Then $Q = \langle \varphi(g_1), \dots, \varphi(g_m) \rangle \leq \varphi(N_0) \leq \varphi(N) \leq Q$, so $Q = \varphi(N_0) = \langle \varphi(h_1), \varphi(h_2) \rangle$ and $\varphi(N) = Q$. The quotient is torsion-free by Theorem 5.1 and finitely presented because $\ker\varphi$ is normally generated by m elements. □

Fournier-Facio constructs a finitely presented torsion-free group G_0 with property (T), a subgroup $\Gamma \leq G_0$ with property (T), an element $t \in G_0$ with $t\Gamma t^{-1} \leq \Gamma$, and a subgroup $J \leq G_0$ isomorphic to a finitely presented infinite simple group, such that $[\Gamma, J] = 1$ and $tJt^{-1} \leq \Gamma$ [FF, §2]. The group G_0 is obtained there as a common quotient of two finitely generated acylindrically hyperbolic groups by Hull's theorem [Hu, Corollary 7.4], which allows the quotient to be chosen acylindrically hyperbolic; we take G_0 to be such a quotient. Put $S = tJt^{-1}$. Since J is simple and nonabelian, J and S are perfect.

Lemma 5.3. *For every homomorphism $\rho\colon G_0 \to \bar{G}$,*

$$\rho(S) \leq \mathfrak{D}_{\bar{G}}(\rho(\Gamma)).$$

Proof. Write $\bar{\Gamma} = \rho(\Gamma)$, $\bar{t} = \rho(t)$, $\bar{J} = \rho(J)$, and $\bar{S} = \rho(S) = \bar{t}\bar{J}\bar{t}^{-1}$. Then $\bar{t}\bar{\Gamma}\bar{t}^{-1} \leq \bar{\Gamma}$, $\bar{J}$ centralizes $\bar{\Gamma}$, and $\bar{S} \leq \bar{\Gamma}$. For $c \in \bar{J}$ and $\ell \in \bar{S}$, the commutator $[\bar{t}c\bar{t}^{-1}, \ell]$ lies in $\mathfrak{D}_{\bar{G}}(\bar{\Gamma})$ by (1). As c ranges over $\bar{J}$, the element $\bar{t}c\bar{t}^{-1}$ ranges over $\bar{S}$, so $[\bar{S}, \bar{S}] \leq \mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. The group $\bar{S}$ is perfect as a quotient of S, so $\bar{S} \leq \mathfrak{D}_{\bar{G}}(\bar{\Gamma})$. □

Proof of Theorem E. Let N be the normal closure of S in G_0; it is nontrivial because S is. By Lemma 5.2 applied to G_0, N, and $\Omega = \varnothing$, there is a surjective homomorphism $\varphi\colon G_0 \to Q$ with Q two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, and $\varphi(N) = Q$. The group Q is infinite, being acylindrically hyperbolic, and it has property (T) as a quotient of G_0.

By Lemma 5.3 applied to φ, $\varphi(S) \leq \mathfrak{D}_Q(\varphi(\Gamma))$. The subgroup $\mathfrak{D}_Q(\varphi(\Gamma))$ is normal in Q, and the normal closure of $\varphi(S)$ is $\varphi(N) = Q$, so

$$\mathfrak{D}_Q(\varphi(\Gamma)) = Q.$$

Both Q and $\varphi(\Gamma)$ have property (T), as quotients of G_0 and Γ, respectively. By the last assertion of Theorem A, every homomorphism from Q to an MF group is trivial. A homomorphism from a quotient of Q composes to one from Q, so no nontrivial quotient of Q is MF. □

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